



Chapter 18 — UI/UX Design Specification



This chapter closes the design gap: screen inventory, layout system, design tokens, key screen wireframes, interaction patterns, and accessibility rules — all producible with **100% free tools** (Penpot or Figma free tier, Excalidraw, open-source fonts and icons).

18.1 Design Philosophy

1. **Dark ops aesthetic, light cognitive load.** Command centers use dark UIs for long-session eye comfort and to make colored signals (severity, alerts) pop. Dark is the default; everything meaningful is communicated by *color + shape + position*, never color alone.
2. **Glanceability over density.** An operator must grasp “is anything wrong?” in under 3 seconds from across the room. Critical state lives in the top status bar and map; detail is one click away, never zero clicks (avoid wall-of-numbers dashboards).
3. **The map is the home.** Every workflow starts from or returns to the map. Panels slide over it; they never replace it.
4. **Real-time must *feel* real-time.** Every live element animates its arrival (pulse, glide, count-up) — motion is information.
5. **AI is visually distinct.** All AI-generated content sits in violet-accented containers with the ⚠️ verify label — instantly distinguishable from ground-truth data.

18.2 Free Design Toolchain

Tool	Use	Cost/License
Excalidraw	Low-fi wireframes, flow sketches	Free, MIT, works offline

Tool	Use	Cost/License
Penpot (self-hosted or SaaS)	Hi-fi mockups, components, prototypes	Free, MPL-2.0 open source
Figma free tier (alternative)	Same, if preferred	Free tier: 3 files — sufficient
Tailwind config	The <i>source of truth</i> for tokens	Free, MIT
Lucide icons	Full icon set	Free, ISC
Inter + JetBrains Mono fonts	UI + data/mono type	Free, SIL OFL

Workflow: sketch in Excalidraw → refine key screens in Penpot → encode tokens directly in `tailwind.config.ts` (design tokens live in code; mockups are disposable, the config is permanent).

18.3 Design Tokens

```
// tailwind.config.ts (excerpt) – the canonical design system
colors: {
  bg:      { base: "#0B0F14", raised: "#121821", overlay: "#1A2230" },
  border:  { subtle: "#232D3D", strong: "#3A4B63" },
  text:    { primary: "#E6EDF3", secondary: "#9FB0C3", muted: "#5D7290" },
  accent:  { DEFAULT: "#3B9EFF" }, // interactive elements
  ai:      { DEFAULT: "#8B7CF6" }, // ALL AI-generated content
  severity: {
    critical: "#FF4D4F", // + diamond shape marker
    high:     "#FF9F1C", // + triangle
    medium:   "#FFD166", // + square
    low:      "#4ADE80", // + circle
    info:     "#38BDF8",
  },
  unit: { friendly: "#4ADE80", drone: "#38BDF8", logistics: "#FACC15" },
}
```

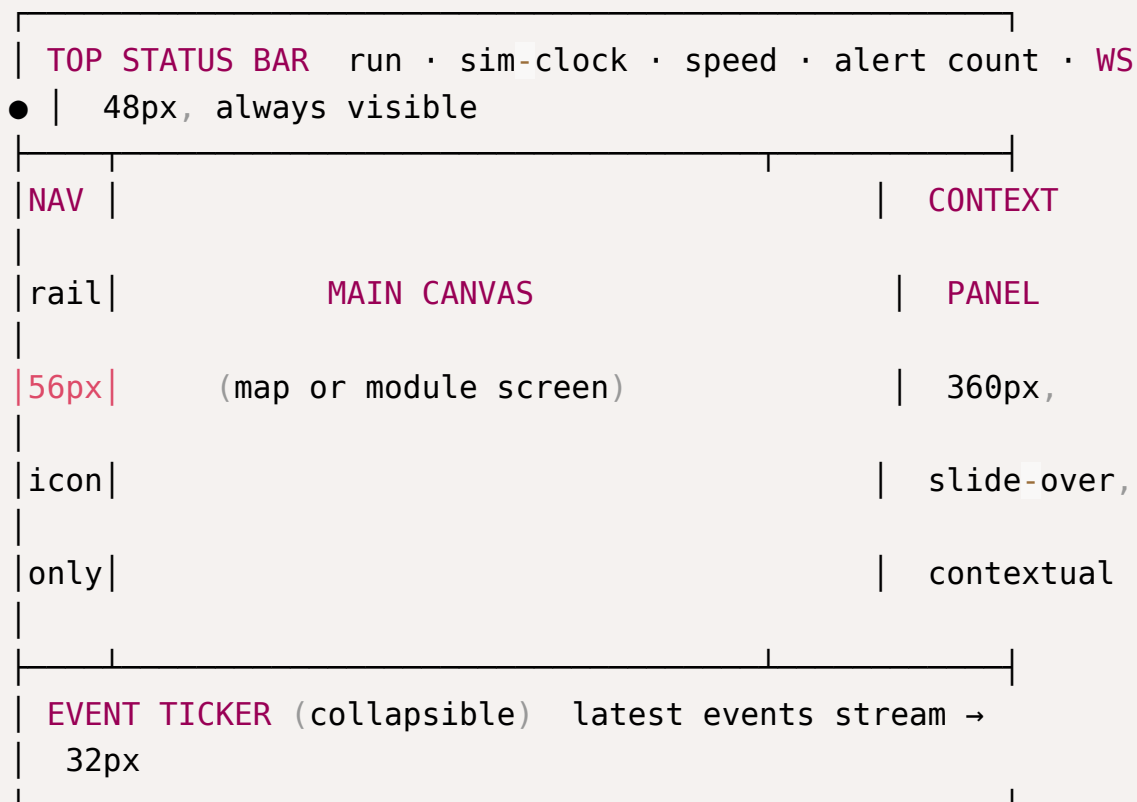
```
fontFamily: { sans: ["Inter"], mono: ["JetBrains Mono"] } //
numbers/coords/IDs always mono
```

Rules: severity colors are **reserved** — never used decoratively. Spacing on a 4 px scale; radii 6/10 px; elevation via border + subtle shadow (no heavy glows).

18.4 Screen Inventory

#	Screen	Route	Primary role	Phase
S1	Login	/login	all	1
S2	Command Dashboard	/	operator+	1
S3	Live Map (full-bleed)	/map	operator+	1
S4	Run Control	/runs	planner+	1
S5	Mission Planner	/missions	planner+	2
S6	Replay / Debrief	/runs/:id/replay	all	2
S7	AI Panel (sitreps + chat)	docked drawer	operator+	3
S8	What-If Theater	/whatif	planner+	3.5
S9	Vision Review	/vision	operator+	4
S10	Logistics Dashboard	/logistics	operator+	5
S11	Personnel Health	/personnel	operator+	5
S12	Analytics & After-Action	/analytics	all	6
S13	Preparedness Advisor	/advisor	planner+	6.5+
S14	War-Room Session	/sessions/:id	session roles	15*
S15	Admin (users, scenarios, corpus)	/admin	admin	1+

18.5 App Shell & Layout System

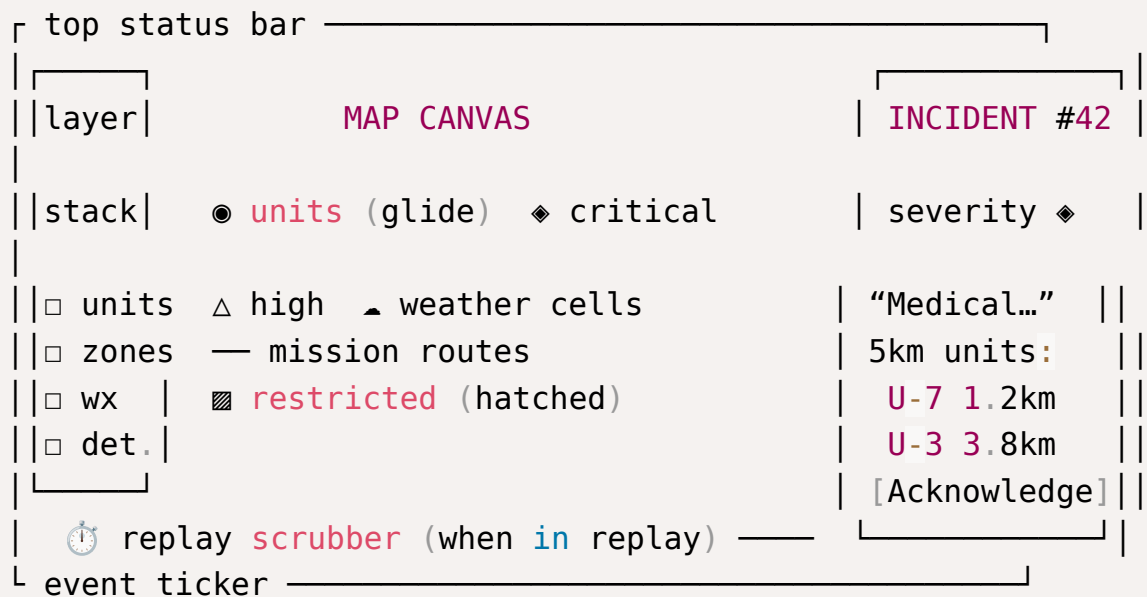


- **Status bar** carries global truth: sim clock (mono font), run state, speed, unacked-alert badge, WebSocket health dot (green/amber pulse/red).
- **Context panel** is the universal detail surface: click a unit → unit card; incident → incident detail; mission → timeline. One consistent place, stackable (breadcrumb back).
- Dashboard (S2) uses a 12-column grid of widgets: mini-map (spans 6), event feed (3), alerts (3), logistics/vitals sparkline rows (Phase 5+).
- Breakpoints: desktop-first (≥ 1280 primary); ≥ 768 collapses context panel to overlay; mobile is read-only PWA territory (future).

18.6 Screen Wireframes — All 15 Screens

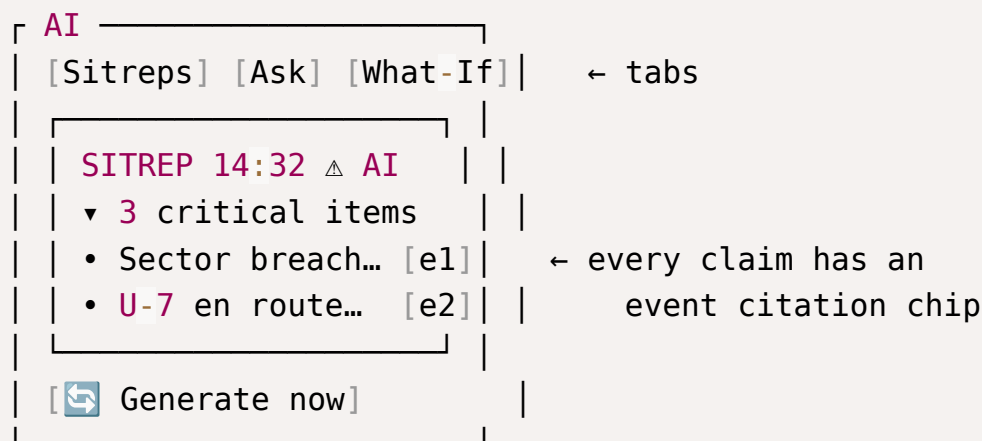
Flagship surfaces first (S3 live map, S7 AI panel), then the remaining screens in numeric order. Wireframes are schematic — spacing is indicative, not pixel-accurate; all screens sit inside the app shell (§18.5) unless noted.

S3 — Live Map (the flagship screen)



Marker language: circles = units (color by type), severity shapes for incidents (◆△■●), dashed outline = stale data (>30 s), pulsing halo = new/unacked. Smooth 4 Hz interpolated motion; click anything → context panel.

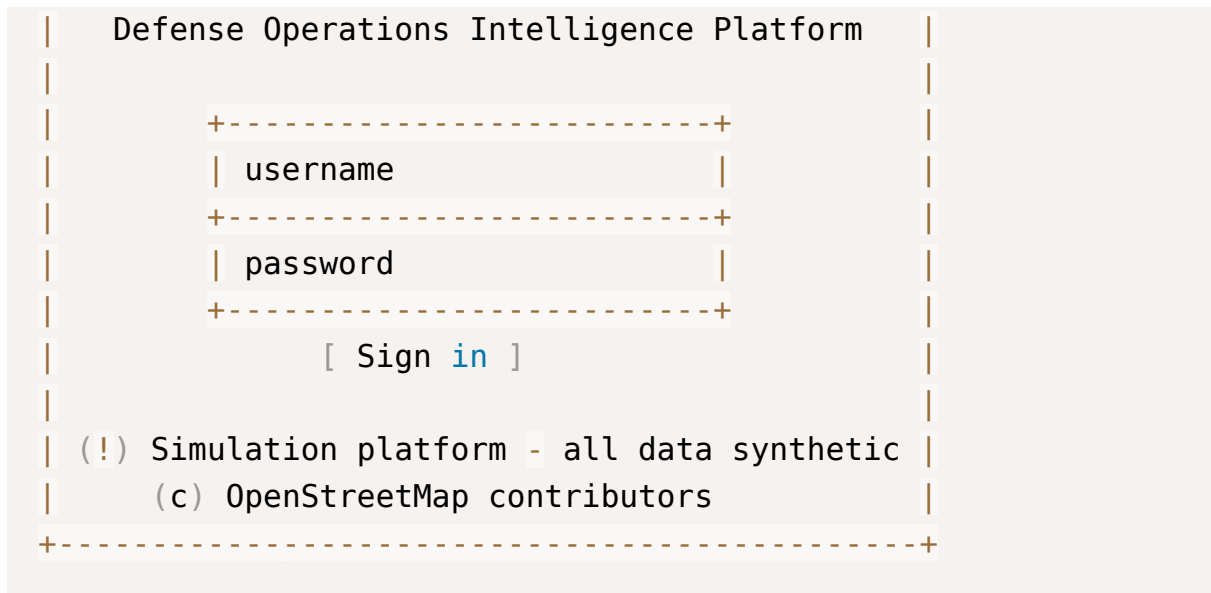
S7 — AI Panel (docked right drawer, violet accent)



Citation chips ([e1]) click through to the actual event, pulsing it on the map — the grounding guarantee made visible.

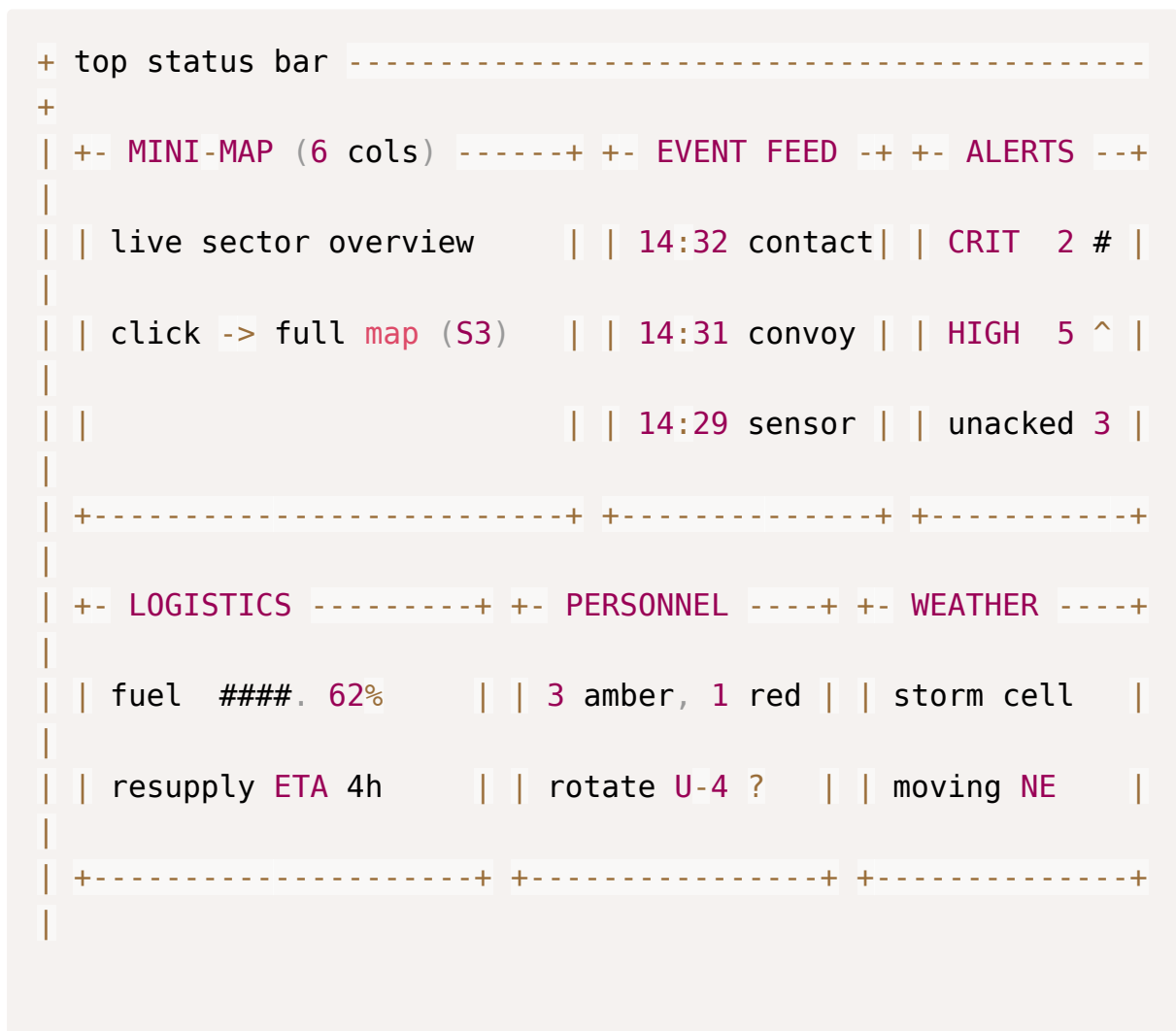
S1 — Login (no app shell)





Dark background, single card, disclaimer and OSM attribution always visible (Ch. 19 §19.4).

S2 — Command Dashboard



```
+ event ticker -----
-+
```

12-column widget grid; every tile links to its module screen; alert tile counts are click-to-filter.

S4 — Run Control

```
+ RUNS -----
-+
| [ + New run ]          filter: [ active | completed | all ]
|
| +-----
+|
| | RUN-031  SC-2 Border Surveillance      * RUNNING
| |
| | seed 8841 | t+02:14 | 4x  [pause] [stop] [open map]
| |
| +-----
+|
| | RUN-030  SC-4 Convoy Escort            / COMPLETE
| |
| | [replay] [after-action] [clone with new seed]
| |
| +-----
+|
| New run:  scenario v | seed [auto] | speed v | [ START ]
|
+-----
--+
```

Seed is always visible — determinism is a first-class UI concept (clone-with-same-seed reproduces a run exactly).

S5 — Mission Planner

```
+ top status bar -----
+
| +- MAP (draw mode) -----+ +- PLAN -----+
|
```

```

| | * waypoints [] survey zone | | Mission: Recon-7 |
|
| | .... naive route (grey dash) | | unit: UAV-2 v |
|
| | ==== optimized route (accent) | | waypoints: 6 |
|
| | #### restricted ~ weather | | dist 14.2km | 38min |
|
| | | | |
| | | | + FEASIBILITY -----+ |
|
| | | | ok battery 71>22% | |
|
| | | | ok avoids R-1 | |
|
| | | | +-----+ |
|
| | | | [optimize] [LAUNCH] |
|
+-----+ +-----+
+-----+
--+
```

Naive-vs-optimized comparison always rendered; LAUNCH stays disabled until the feasibility card is fully green, and infeasible verdicts list *why*.

S6 — Replay / Debrief

```

+ REPLAY: RUN-030 -----
-+
| +- MAP (state at scrub time) -----+ +- TIMELINE -----+
|
| | entities at t | | 00:00 run start |
|
| | war-room annotations replay | | 00:41 # contact |
|
| | RTA inject markers | | 01:12 ^ inject (RTA) |
|
| | | | 02:03 medevac done |
|
```

```

| +-----+ +-----+
|
| <<  ---o----- >>  1x 4x 8x  [jump to e
v]|
| score timeline _.-=#  |  [export after-action report]
|
+-----+
--+
```

Everything evented appears here at its original sim-time: annotations, injects, decisions — the deterministic replay guarantee made visible.

S8 — What-If Theater

```

+ WHAT-IF THEATER -----
-+
| Prompt: "storm hits sector C at 14:00 during convoy run"
|
| +- AI SCENARIO DRAFT (violet, verify label) -----+
|
| | + weather: storm cell, sector C @ 14:00 |
|
| | + incident: road blocked, route R-2 @ 14:20 |
|
| | [edit actions]  [approve & run]  [discard] |
|
| +-----+
+|
| headless run: #####.... 74%
|
| +- ANIMATED PLAYBACK -----+ narration (AI, cited)
|
| | condensed 60s map animation | "At 14:06 the convo
y..."|
| +-----+ [compare vs baseline]
|
+-----+
--+
```

The approve/edit gate between draft and run is mandatory (Ch. 13 FR) — the UI makes skipping it impossible.

S9 — Vision Review

```
+ VISION -----
-+
| +- FRAME VIEWER -----+ +- DETECTIONS -----+
|
| | [prerecorded aerial frame] | | vehicle 0.91 > map |
|
| | +----+ vehicle | | person 0.84 > map |
|
| | +----+ +----+ person | | smoke 0.67 ! rev |
|
| | +----+ | | frame risk: HIGH |
|
| +-----+ +-----+
|
| job: recon-04.mp4 | 312/900 frames | [accept all >= 0.80]
|
+-----
--+
```

Low-confidence detections queue for human review; accepted detections become map events with model-version lineage.

S10 — Logistics Dashboard

```
+ LOGISTICS -----
-+
| fuel ###.. 58% v | medical ####. 77% | rations ##... 41% v
|
| +- BURN-DOWN + FORECAST -----+ +- ALERTS -----+
|
| | stock ----\ | | ! fuel depot D-2 |
|
| | actual \____ forecast | | empty in ~6h |
|
| | \ . . . . | | ! medkits at post P-4 |
```

```

|
| +-----+ +-----+
|
| depot table: depot | stock | consumers | resupply ETA
|
+-----+
--+
```

Forecast line (dotted) comes from the analytics service with uncertainty band; alert thresholds are admin-configurable.

S11 — Personnel Health

```

+ PERSONNEL -----
-+
| roster 42 | ok 36 | amber 5 | red 1
|
| +- VITALS GRID -----+ +- DETAIL: S-114 -----+
|
| | id    hr    fatigue    state | | hr spark _.-=-. |
|
| | S-114 128  ####.      RED   | | fatigue rising 4h |
|
| | S-021  96  ##...      ok    | | suggest: rotate unit |
|
| | S-007  88  #....      ok    | | [create task > S14] |
|
| +-----+ +-----+
|
| threshold crossings -> alerts, map badge, ticker
|
+-----+
--+
```

S12 — Analytics & After-Action

```

+ ANALYTICS -----
-+
| run v | compare: [ RUN-030 ] vs [ RUN-027 ]
```

```

|
| +- KPIs -----+ +- RESPONSE TIMES -+ +- EFFICIENCY -----+
|
| | p95 resp    | | histogram _#=_    | | used vs wasted    |
|
| | 12.4 min    | | by severity          | | by category          |
|
| +-----+ +-----+ +-----+
|
| decision timeline | AI narrative (cited) | [export report]
|
+-----+
--+
```

Compare mode is the training payoff: same scenario + seed, two runs, side-by-side deltas.

S13 — Preparedness Advisor (Ch. 14)

```

+ ADVISOR -----
-+
| inventory v3 | budget: X Cr | [ run scenario battery ]
|
| +- GAP REGISTER -----+ +- PORTFOLIOS -----+
|
| | 1 # medevac lift capacity    | | A balanced 0.92X    |
|
| | evidence: RUN-101,104 >      | | B mobility 0.97X    |
|
| | 2 ^ radios per patrol        | | delta p95 -18% +-6% |
|
| +-----+ +-----+
|
| historical parallels: Nepal 2015 airlift > (cited, factual)
|
| (!) simulation-derived decision support - not procurement
|
+-----+
--+
```

Every number click-throughs to its evidence runs; documented fact and simulated projection use different visual styling (Ch. 14 §14.6).

S14 — War-Room Session (Ch. 15)

```
+ SESSION: EX-DRISHTI | role: COMMANDER -----
-+
| +- SHARED MAP -----+ +- TEAM -----+
|
| | sector grid A / B / C / D      | | * Cdr   you      |
|
| | * annotations (named cursors) | | * Op-A  sector A  |
|
| | ^ inject markers (RTA)        | | * Op-B  sector B  |
|
| |                               | | o Observer      |
|
| +-----+ +-----+
|
| +- TASK BOARD -----
+|
| | QRF > A  [ack]   medevac > B  [doing]   recon > C [open]
| |
| +-----+
+|
+-----+
--+
```

Operators see only their sector's streams (server-side filtering); the commander sees all sectors plus the task board.

S15 — Admin

```
+ ADMIN -----
-+
| [Users] [Scenarios] [Corpus] [Catalog] [Audit]
|
| Users:      name | role v (admin/planner/operator/viewer)
|
| Scenarios:  YAML editor + validate ok/err + [test headless]
```

```

| Corpus:      upload SOP/REF/POL -> chunk -> embed ###..
|
| Catalog:     equipment items (Ch. 14) cost | effects | upkeep
|
| Audit:       who | what | when (append-only, exportable)
|
+-----+
--+
```

Scenario validation reuses the same Validation Gate as Ch. 13 — one gate, three authors (human, LLM, RTA).

18.7 Component Library

Base: **shadcn/ui** (free, MIT, copy-in components) + Tailwind. DOIP-specific components to build once and reuse:

`<StatusDot>` (WS/health) · `<SeverityBadge>` (shape+color) · `<SimClock>` (mono, speed-aware) · `<EventRow>` (feed/ticker) · `<MetricTile>` (dashboard) · `<AICard>` (violet frame + verify label + citation chips) · `<MapMarker>` set · `<ReplayScrubber>` · `<FeasibilityVerdict>` · `<SparkTile>` (vitals/logistics)

18.8 Interaction Patterns

Pattern	Rule
Live arrival	New feed items slide in; markers pulse once; counters count-up. Never teleport values.
Alert escalation	Critical → toast + status-bar badge + map ping. Toasts never block the map; max 3 stacked, then "+N more".
Optimistic UI	Acks/tasks apply instantly with pending dot; reconcile on event confirmation.
Loading / empty / error	Every widget defines all three states (skeleton shimmer / helpful empty copy / retry action). Enforced by the Definition of Done (Ch. 9).
Disconnect	Status dot amber-pulses; banner "Reconnecting — you may be seeing stale data"; map dims 10%. On resume, missed events replay visually at 8x.
Keyboard	<code>Space</code> pause/resume (planner+), <code>F</code> follow-mode, <code>E</code> event feed focus, <code>1–9</code> layer toggles, <code>?</code> shortcut sheet.

18.9 Accessibility

- Contrast $\geq$ WCAG AA on dark backgrounds (all token pairs pre-checked).
- **Never color-only:** severity always pairs color with shape (◆△■●) and label — also solves the red/green colorblind case.
- Full keyboard navigability for non-map surfaces; map interactions have panel-based equivalents (incident list = map markers).
- Motion-reduction media query: disables pulses/glides, keeps state changes.
- Live regions (`aria-live="polite"`) for feed updates; `assertive` only for critical alerts.

18.10 Design Debt Policy

Design work is timeboxed *inside* each phase (no separate design phase): wireframe the phase's screens in sprint 1 of the phase, refine only what the demo needs. Penpot files live in `docs/design/` ; exported PNGs of final screens are committed alongside for zero-tooling review.